IT Minds Institute Website: Design & Technical Specification

Version 1.1 (Draft for internal review) · Updated October 2026 Status: Discovery and design baseline · **Stack:** React, Supabase, Vercel, GitHub

1. Purpose and Scope

This document sets the baseline for the design, architecture and delivery of the IT Minds Institute website (www.itminds.ac.za). It builds on the material received from the client: the Cornerstone document (mission, vision, values and the Learn, Innovate, Empower cycle), the institutional overview and programme structure, the cycle graphic, the logo, and the covering brief.

The website must do two jobs equally well:

1. Present the institution's **academic offering** clearly and credibly: six programme areas and 24 QCTO and NQF-aligned programmes, with the QCTO and SAQA accreditation visible.
2. Position the institute as **future-focused** through a prominent **Skills & Training** section (in partnership with PATRIDOS), a **Business School**, and a highly visible **E-Learning** entry point.

Out of scope for this phase: student information system integration, payment processing, native mobile applications, and the LMS itself (the website links to it).

2. Client Requirements Summary

#	Requirement	Source	Priority
R1	Faculties and programmes structured per the institutional overview (six areas, 24 programmes)	Overview	Must
R2	Dedicated **Skills & Training** tab in main navigation, in partnership with PATRIDOS	Brief	Must
R3	Ten corporate short courses showcased (see Section 5.5)	Brief	Must

#	Requirement	Source	Priority
R4	Emerging-technology and AI-related programmes within Skills & Training	Brief	Must
R5	**Business School** tab (course content to follow)	Brief	Must
R6	**E-Learning** button/tab visible on every page	Brief	Must
R7	PATRIDOS co-branding (logo to be supplied)	Brief	Must
R8	Modern, professional, future-focused presentation	Brief	Must
R9	QCTO Skills Development Provider status (SDP122596) and SAQA/NQF alignment displayed accurately	Overview	Must
R10	Mission, vision, values and the Learn, Innovate, Empower cycle expressed in the design	Cornerstone	Must
R11	Student support services presented (academic support, careers, labs, internships, entrepreneurship, wellness)	Overview	Should

3. Information Architecture

3.1 Primary navigation

- **About** (Institute, Mission and Vision, Our Cycle, Governance, Accreditation, Student Support, Contact)
- **Faculties** (six faculty landing pages, programme detail pages beneath)
- **Skills & Training** (in partnership with PATRIDOS)
- **Business School**
- **Admissions** (How to apply, Fees, Key dates, Enquiry form)
- **News & Events**
- **E-Learning** (persistent call-to-action button, styled distinctly from text links)

3.2 Sitemap

```
/
├── about
│   ├── mission-vision
│   ├── our-cycle             (Learn, Innovate, Empower)
│   ├── governance
│   ├── accreditation         (QCTO, SAQA, NQF)
│   └── student-support
├── faculties
│   ├── software-development
│   ├── networking-cloud
│   ├── data-ai
│   ├── it-support-systems
│   ├── digital-design-multimedia
│   └── emerging-tech-innovation
│       └── [programme-slug]
├── skills-training
│   ├── corporate-training
│   │   └── [course-slug]
│   ├── ai-and-emerging-tech
│   │   └── [programme-slug]
│   └── partnership           (PATRIDOS)
├── business-school
│   └── [course-slug]         (placeholder until content is received)
├── admissions
├── news
├── contact
└── e-learning                (external LMS link or gateway page)
```

3.3 Skills & Training structure

The section opens with a short introduction and the PATRIDOS partnership statement, then splits into two clear tracks so visitors self-select:

- **Corporate Short Courses:** role- and sector-based training for organisations.
- **AI & Emerging Technology Programmes:** fundamentals, workplace tools, generative AI, business and productivity applications, and related technology skills.

Separating accredited programmes from short courses. Faculty programmes are the NQF-aligned qualifications. Skills & Training holds the short, non-credit-bearing courses. Where topics overlap (for example AI & Machine Learning Fundamentals and AI Prompt

Engineering), each page links to the other and states clearly which is the accredited qualification and which is the short course.

4. Design Direction

4.1 Principles

- **Credible first.** Academic institutions are judged on trust. Clean layouts, generous whitespace, restrained colour.
- **Future-focused, not flashy.** Modern typography, subtle motion, sharp geometry. No gimmicks.
- **Two audiences, one site.** Prospective students need programme clarity. Corporate buyers need quick course scanning and an enquiry route.
- **Mobile-first.** The majority of local traffic is expected on mobile devices.

4.2 Visual system (to be finalised against the institution's brand assets)

Element	Direction
Brand green	Approximately #3F9A12, sampled from the logo; final value to be confirmed from a vector logo
Brand slate	Approximately #2A3848, used for headings, header and footer
Neutrals	White and warm off-white backgrounds, slate text
Faculty accents	Six accents mirroring the institutional overview: green (Software Development), blue (Networking & Cloud), purple (Data & AI), orange (IT Support & Systems), teal (Digital Design & Multimedia), crimson (Emerging Tech & Innovation). Used as fills, borders and tags only; text colours are checked against contrast rules
Cycle colours	Blue (Learn), green (Innovate), orange (Empower), reserved for the cycle graphic
E-Learning	One distinct, high-contrast button colour used nowhere else

Element	Direction
Headings	A bold geometric sans-serif, close to the weight of the logo wordmark
Body	A highly legible humanist sans-serif, 16 to 18 px base
Imagery	Real campus, lab and student photography preferred; consistent treatment for stock
Iconography	Single line-icon family, 1.5 px stroke, matching the icons in the overview
Motion	Short fade and slide transitions (150 to 250 ms); respects reduced-motion settings

4.3 Core components

- Global header with sticky navigation and a persistent **E-Learning** button
- Hero carrying the tagline *Learn | Innovate | Empower* with primary and secondary calls-to-action
- **Cycle section:** three-stage Learn, Innovate, Empower graphic rebuilt as native components
- **Faculty cards** colour-coded by area, with programme count and NQF range
- Programme listing with filters (faculty, NQF level, mode)
- Programme detail template (overview, NQF level, entry requirements, curriculum, career outcomes, apply)
- **Accreditation strip:** QCTO Skills Development Provider (SDP122596), SAQA and NQF alignment
- Course card (title, audience, duration, delivery mode, enquire)
- Partner banner (institute and PATRIDOS lock-up)
- Student support grid (academic support, career guidance and placement, labs, e-learning, internships, entrepreneurship, wellness)
- Enquiry and application-interest forms
- Footer with accreditation, contact, social links and policy links

4.4 Page templates

1. **Home:** hero, four pathways (Study, Train, Business School, E-Learning), Learn-Innovate-Empower cycle, six faculties, Skills & Training spotlight, accreditation strip, partner strip, news, enquiry strip.
2. **About:** purpose, mission and vision, values, cycle, governance structure, accreditation, student support.
3. **Faculty landing:** introduction, programmes grouped by NQF level, faculty contact.
4. **Programme detail:** structured content blocks driven by data.
5. **Skills & Training hub:** two-track layout, course grid, PATRIDOS partnership section, corporate enquiry form.
6. **Course detail:** outcomes, audience, duration, delivery format, enquiry.
7. **Business School:** introduction with a "courses announced shortly" state and an interest-registration form until content is supplied.

4.5 Accessibility and inclusion

WCAG 2.2 AA target: colour contrast, keyboard navigation, visible focus states, semantic headings, alt text, and form error clarity.

5. Content Model

5.1 Faculties and programmes

The institutional overview defines six faculties and 24 programmes, every one marked as AI-integrated and aligned to the NQF and QCTO occupational qualifications. Each programme carries the same fields so templates stay consistent: name, NQF level, qualification type, duration, mode of study, entry requirements, summary, curriculum highlights, career outcomes and SAQA identifier where registered.

Faculty	Programme	NQF level
Software Development	Software Development	4
Software Development	Web Development	4
Software Development	Mobile App Development	5
Software Development	Full Stack Development	5

Faculty	Programme	NQF level
Networking & Cloud Computing	Network Support	4
Networking & Cloud Computing	Cloud Computing	5
Networking & Cloud Computing	Cybersecurity	5
Networking & Cloud Computing	Systems Administration	5
Data & AI Technologies	Data Analytics	5
Data & AI Technologies	AI & Machine Learning Fundamentals	5
Data & AI Technologies	Big Data Technologies	5
Data & AI Technologies	Business Intelligence	5
IT Support & Systems	IT Support Technician	4
IT Support & Systems	Computer Hardware	4
IT Support & Systems	Operating Systems	4
IT Support & Systems	IT Service Management	5
Digital Design & Multimedia	UI/UX Design	4
Digital Design & Multimedia	Graphic & Motion Design	5
Digital Design & Multimedia	Digital Content Creation	4
Digital Design & Multimedia	Video Editing & Production	5
Emerging Tech & Innovation	Internet of Things (IoT)	5
Emerging Tech & Innovation	Robotic Process Automation (RPA)	5
Emerging Tech & Innovation	Blockchain Technology	5
Emerging Tech & Innovation	AI Prompt Engineering	5

The overview calls these "programmes and courses" and the brief calls the groupings "faculties". The site uses **Faculties** in navigation until the client confirms otherwise.

5.2 Content ownership

All editable content is held in the database and edited through a protected admin area, so the client can update programmes and courses without a code release.

5.3 Placeholders and pending assets

Item	Status	Owner
Logo (PNG, 1254 × 1254, white background)	Received. A transparent vector (SVG) and a horizontal lock-up are needed for the header and favicon	Client
Cornerstone document (mission, vision, values, cycle)	Received	Client
Institutional overview and programme graphic	Received	Client
Learn, Innovate, Empower cycle graphic	Received; rebuilt as native components	Client
PATRIDOS logo and usage rules	Awaiting client	Client
Business School course list	Awaiting client	Client
Programme detail (duration, entry requirements, curriculum, fees, SAQA IDs)	Awaiting client	Client
E-Learning platform URL	To confirm	Client
Governance names, leadership profiles	To confirm	Client
Social media links (Facebook, LinkedIn, Instagram)	To confirm	Client
Photography (campus, labs, students)	To confirm	Client

5.4 Institutional identity content

The Cornerstone document supplies the copy for the About section and the home page.

- **Mission:** to provide quality, accessible and industry-relevant education, training and practical skills that empower individuals to succeed in today's digital world.
- **Vision:** to be a leading centre of excellence in digital education, research and innovation, driving digital transformation and empowering individuals to lead in an interconnected and evolving digital world.
- **Core values (Cornerstone document):** Innovation, Excellence, Digital Leadership, Integrity, Lifelong Learning, Collaboration, Adaptability.

- **Operating cycle:** Learn builds the foundation, Innovate turns knowledge into action, Empower creates opportunity for individuals, families, industries and communities.
- **Closing message:** the "Message to Future Generations" is used on the About page and in the footer sign-off.

The overview graphic lists a different set of values (Integrity, Innovation, Excellence, Respect, Accountability, Teamwork Centred). The site will use the Cornerstone list unless the client says otherwise.

5.5 Corporate short courses (launch catalogue)

1. AI in the Modern Business Environment
2. AI Productivity Masterclass
3. AI for Human Resources
4. AI for Finance & Accounting
5. AI for Data Analysis
6. AI for Small Businesses & SMMEs
7. AI-Powered Entrepreneurship & Innovation
8. AI Leadership for Managers and Executives
9. AI for Public Sector & Government
10. AI Literacy for Schools and Young People

Course descriptions, durations and audiences are to be drafted with the client and entered through the admin area.

6. Technical Architecture

6.1 Platform decision: React and framework

The project is a public website where **search visibility, load speed and shareable links** matter. We recommend building with **React using Next.js**, deployed on Vercel. This gives server-rendered pages, strong SEO, image optimisation and straightforward Vercel deployment.

On React Native: React Native targets native mobile apps. It is not the right foundation for the public website. Where the team wants to share code with a future mobile app, we can structure shared logic (types, API clients, validation) in a separate package and use

React Native for the app later. This decision should be confirmed with the client before development starts.

6.2 Components

Layer	Choice	Notes
Front end	Next.js (App Router), React, TypeScript	Static generation with revalidation for content pages
Styling	Tailwind CSS with design tokens	Tokens map to the visual system in Section 4.2
Data and auth	Supabase (PostgreSQL, Auth, Storage)	Row Level Security on all tables
Hosting	Vercel	Preview deployment per pull request
Source control	GitHub	Protected main branch, pull request reviews
Email	Transactional provider (SMTP/API)	Enquiry notifications
Analytics	Privacy-respecting analytics	Consent-aware

6.3 Data model (initial)

Table	Key fields
faculties	id, slug, name, summary, accent_colour, icon, hero_image, sort_order, published
programmes	id, faculty_id, slug, name, nqf_level, qualification_type, saqa_id, credits, duration, mode, entry_requirements, curriculum, outcomes, ai_integrated, published
accreditations	id, body (QCTO, SAQA), reference_number, description, display_order
training_tracks	id, slug, name, description (corporate, ai_emerging_tech)

Table	Key fields
training_courses	id, track_id, slug, name, audience, duration, delivery_mode, summary, outcomes, partner (PATRIDOS flag), published
business_school_courses	id, slug, name, summary, details, published
student_support_services	id, name, description, icon, sort_order
enquiries	id, type, name, email, phone, organisation, message, source_page, created_at, status
news_posts	id, slug, title, body, cover_image, published_at
site_settings	key, value (e-learning URL, contact details, social links, partner assets)
profiles	id, role (admin, editor)

6.4 Security and privacy

- Row Level Security: public read for published content only; writes limited to authenticated editors and admins.
- Enquiry inserts allowed for anonymous users through a validated server route with rate limiting and bot protection.
- Secrets only in Vercel and Supabase environment settings, never in the repository.
- Compliance with **POPIA**: consent text on forms, defined retention period for enquiries, privacy policy page.
- Security headers (CSP, HSTS, frame protection) configured at the Vercel layer.

6.5 Performance and SEO targets

- Lighthouse 90+ across performance, accessibility, best practices and SEO on key templates
- LCP under 2.5 s on mid-range mobile over 4G
- Structured data (Organization, Course, EducationalOrganization), XML sitemap, clean canonical URLs, Open Graph images

7. Team Workflow

7.1 Environments

Environment	Purpose	Source
Local	Development in VS Code with a local Supabase instance	Feature branches
Preview	Per-PR review by team and client	Vercel preview URL
Staging	Client sign-off	`develop` branch
Production	Live site	`main` branch

7.2 Git conventions

- Branch names: `feature/`, `fix/`, `chore/` followed by a short description
- Conventional commit messages (`feat:`, `fix:`, `docs:`, `refactor:`)
- Pull requests require at least one reviewer and a passing pipeline
- Database changes only through versioned Supabase migrations committed to the repository

7.3 Quality gates (GitHub Actions)

Lint, type check, unit tests, build, and accessibility checks on every pull request. Production deploys only from `main`.

7.4 Suggested repository layout

```
/app                routes and page templates
/components         UI components
/lib                data access, validation, utilities
/supabase/migrations versioned schema changes
/docs               this specification, decisions log
/public             static assets
```

7.5 VS Code setup

Shared workspace settings with ESLint, Prettier, TypeScript and Tailwind IntelliSense extensions recommended through `.vscode/extensions.json`, so formatting is identical across the team.

8. Delivery Plan

Phase	Focus	Outcome
0. Discovery	Review documents, confirm open items	Signed-off scope and sitemap
1. Design	Wireframes, visual system, key templates	Approved designs for Home, Faculty, Programme, Skills & Training
2. Foundation	Repository, CI, Supabase schema, design tokens, layout	Working skeleton on preview URL
3. Build	Templates, content import, admin area, forms	Feature-complete staging site
4. Content and QA	Content load, accessibility, performance, security review	Release candidate
5. Launch	DNS, redirects, monitoring, handover	Production release

Phase durations will be confirmed after the open items in Section 9 are resolved.

9. Open Items for Client Clarification

1. Confirm the labels: "Faculties" in the brief against "programmes and courses" in the overview.
2. Which programmes are registered QCTO occupational qualifications, and which are institute-designed courses? Provide SAQA IDs where registered.
3. Values: use the seven in the Cornerstone document or the six in the overview graphic?
4. Vector logo (SVG) and brand guide with approved colour values.
5. PATRIDOS logo, partnership wording, and any approval process for how it is displayed.

6. E-Learning: which platform, and should the button open in a new tab or pass through a gateway page?
7. Business School: launch with a registration-of-interest page, or hold the tab until courses are ready?
8. Corporate courses: pricing visible or enquiry-only; in-person, online or blended delivery.
9. Application process: link to an existing portal, or a simple interest form for this phase?
10. Programme detail for all 24 programmes: duration, entry requirements, curriculum, fees.
11. Domain and hosting ownership, social media handles, and who administers content after launch.

10. Risks and Mitigations

Risk	Impact	Mitigation
Content supplied late	Delays testing and launch	Placeholder states and data-driven templates, so launch is not blocked
Accreditation claims overstated	Regulatory and reputational exposure	Show the QCTO SDP number and per-programme SAQA IDs only as the client can evidence them; wording signed off by the client
Brand assets incomplete (raster logo only)	Rework of header, favicon and visual system	Tokenised design system; request SVG early
Two value sets in the source documents	Inconsistent messaging	Confirm one list before copy is loaded
Scope growth (portal, payments)	Budget and timeline	Change-request process; features logged for a later phase
Framework mismatch (React Native for web)	Poor SEO and performance	Confirm Next.js decision in Phase 0

11. Approval

Role	Name	Date
Project lead		
Client representative		